

Extended Abstract

Motivation Fine-tuning large language models (LLMs) is increasingly expensive, both in terms of computational resources and labeled data. While methods like Direct Preference Optimization (DPO) have shown promise, they rely on techniques such as online sampling or policy gradients, which can be inefficient and unstable. This becomes a problem when generating new samples is costly or impractical. To address this, we explore Implicit Q-Learning (IQL), an off-policy reinforcement learning method. IQL has desirable properties for LLM fine-tuning because it is offline and emphasizes stable policy learning from static datasets. Our goal is to evaluate whether IQL can match or exceed the performance of methods like DPO, while offering improved efficiency through its off-policy nature.

Method We compare three fine-tuning strategies for aligning LLM results with human preferences. The three strategies are supervised fine-tuning, Direct Preference Optimization, and Implicit Q-Learning.

SFT serves as a baseline. We trained it with standard cross-entropy loss on pairs of prompts and responses from the SmolTalk dataset. DPO fine-tunes a model using pairs of preferred and rejected completions from the UltraFeedback dataset, optimizing a loss that compares the log-likelihoods of each response relative to our SFT checkpoint.

Our extension focuses on IQL, which trains a policy based on static preference data without requiring online sampling or exploration. Instead of estimating explicit Q-values or value functions, we approximate the advantage of each action using the difference in log-probabilities between the current policy and the reference model. This gives us a stable and efficient off-policy update rule for fine-tuning.

Implementation We implemented three fine-tuning objectives: SFT, DPO, and IQL. SFT was trained on the SmolTalk dataset using cross-entropy loss. DPO and IQL both used preference data from the UltraFeedback dataset. Throughout training, we logged loss values and periodically evaluated each model by comparing our generated responses with reference responses from the Qwen/Qwen2.5-0.5B-Instruct model. From there, we used the Nemotron reward scoring system to generate scores for both responses. If our model received a better score than the reference model, we counted that as a win.

Results Our experiments showed that Implicit Q-Learning achieved a win rate of 0.59 which nearly matches Direct Preference Optimization’s win rate of 0.61. Both of these outperformed the Supervised Fine-Tuning baseline of 0.48. In terms of training efficiency, IQL had an average time of 0.98 hours per epoch which is much better than DPO’s 1.44 hours per epoch. We also observed in the graphs that the IQL training was more stable and smooth.

Discussion Our results showed that Implicit Q-Learning can nearly match the win rate performance of Direct Preference Optimization while significantly improving the speed/efficiency of its training. Despite its simplicity, IQL consistently reduced the training time per epoch which makes it a good alternative for scenarios where computation or sampling is limited. This suggests that there are situations where off policy methods can be practical for large language model fine-tuning. It is interesting that the drastic increase in efficiency did not have a large performance cost. We did see a very small drop in the win rate but that is significantly smaller than the huge speed boost.

Conclusion Our experiments demonstrate that Implicit Q-Learning is a promising off policy alternative for LLM fine-tuning. While methods like DPO are effective, they depend on computationally intensive processes. In contrast, IQL uses only static preference data and still maintains a very similar win rate performance with much better efficiency. This suggests that IQL may be a viable foundation for future scalable, offline fine-tuning techniques.

Off-Policy Implicit Q-Learning for Efficient Fine-Tuning of LLMs

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Abstract

Fine-tuning large language models with human preferences is a key challenge in modern natural language processing. However, methods like Direct Preference Optimization rely on online sampling, making them expensive and inefficient to train at scale. In this project, we explore Implicit Q-Learning, an off-policy reinforcement learning method, and adapt it for preference-based fine-tuning of large language models.

We compare IQL with two baselines: supervised fine-tuning and Direct Preference Optimization. These models are trained on the SmolTalk and UltraFeedback datasets and are evaluated using the Nemotron reward model, counting a generation as a "win" if its score was greater than the reference response.

Our results show that IQL achieves nearly the same win rate as DPO (0.59 vs 0.61) while reducing the average training time per epoch by more than 30%. We also observe that IQL training is more stable, with smoother loss curves and less fluctuation in evaluation scores. These findings suggest that IQL could be a practical and efficient alternative to LLM fine-tuning, offering comparable performance with significantly reduced computational cost.

1 Introduction

Fine-tuning large language models to align with human preferences is a very important goal in natural language processing. Despite this, the process is expensive, slow, and unstable, especially when using reinforcement learning approaches that rely on online sampling or policy gradients. Methods like Direct Preference Optimization (DPO) have been proven to be effective, but are typically very resource-intensive and quite difficult to scale. In settings where data collection is limited or sampling is expensive, these methods are not very effective.

Recent work in offline reinforcement learning shows a potential solution. Implicit Q-Learning (IQL) in particular allows policy learning from static datasets, which does not require on-policy actions or exploration. IQL bypasses many of the instabilities that come with traditional Q-learning and policy-gradient methods. Despite its efficiency and stability, IQL has seen little application in the context of language models and its performance in aligning LLMs with human preferences is largely unexplored.

In this work, we investigate whether IQL can be effectively applied to preference-based LLM fine-tuning. We compare it with two other approaches: supervised fine-tuning (SFT), which acts as our baseline for general quality, and DPO, a widely used preference optimization method. Our models are evaluated on the SmolTalk and UltraFeedback datasets. Our evaluation focuses on two main questions: Can IQL achieve sufficient alignment performance compared to DPO? and Can IQL improve training efficiency due to its fully off-policy nature?

SFT (used as a baseline): $\max_{\theta} \mathbb{E}_{x,y \in D} \sum_{t=1}^{ y } \log \pi_{\theta}(y_t x, y_{<t})$	DPO (evaluate against SFT): $\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x,y_0,y_1) \sim D} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_1 x)}{\pi_{\text{ref}}(y_1 x)} - \beta \log \frac{\pi_{\theta}(y_0 x)}{\pi_{\text{ref}}(y_0 x)} \right) \right]$	IQL (evaluate against SFT): $\begin{aligned} \hat{V}(s) &\leftarrow \arg \min_{\hat{V}} \mathbb{E}_{(s,a) \sim D} \left[\ell_{\hat{V}}^2 \left(V(s) - \hat{Q}(s,a) \right) \right] \\ \hat{Q}(s,a) &\leftarrow \arg \min_{\hat{Q}} \mathbb{E}_{(s,a,s') \sim D} \left[\left(\hat{Q}(s,a) - (r + \gamma \hat{V}(s')) \right)^2 \right] \\ \pi &\leftarrow \arg \max_{\pi} \mathbb{E}_{(s,a) \sim D} \left[\log_{\pi}(a s) \exp \left(\frac{1}{\alpha} (\hat{Q}(s,a) - \hat{V}(s)) \right) \right] \end{aligned}$
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Figure 1: Method Overview.

2 Related Work

There have been many prior works focusing on improving the stability and efficiency of large language model fine-tuning by using off-policy sampling. One method that worked well in this realm was Tapered Off-Policy REINFORCE (1). This algorithm used a tapered form of importance sampling, which results in an efficient offline learning method for fine-tuning large language models. Not only did this off-policy algorithm show gains in solution generation, but it also showed improvement in training data efficiency, a benchmark that we will be trying to reach with our method.

In another paper, a new reinforcement learning algorithm was introduced, called Any-Generation Reward Optimization (2). The Any-Generation Reward Optimization (AGRO) model was specifically designed for fine-tuning large language models, like we will be doing in this project. This offline method uses the concept of generation consistency, which means that the optimal policy is consistent across any output of the model. AGRO uses this guarantee of consistency to its advantage, helping it outperform on-policy methods. It should be noted, though, that this method falls short when off policy data generation is overused, causing the gradients this algorithm calculates to become unstable.

One final report we saw on offline reinforcement learning methods was a test- and method-agnostic that is used to automatically select the best reinforcement learning policy (4). There are a wide array of performances that different models can achieve due to their hyperparameters and functions, however, this model selects the best policy after training and comparing different policies against each other. This approach helps produce better performing policies when it comes to efficiency and accuracy, and is a great example of the capabilities of offline reinforcement learning.

What we noticed when researching our specific extension is that there is next to nothing reported on Implicit Q-Learning. This could be because the method was only created in 2021 (3) and therefore there has not been enough time for substantial research on the method. We aim to change this by implementing and testing our own method of Implicit Q-Learning, and testing how effective it can be in increasing efficiency versus an on-policy method.

3 Method

Our study evaluates three fine-tuning approaches for aligning large language models with human preferences. As mentioned previously, these methods are Supervised Fine-Tuning, Direct Preference Optimization, and Implicit Q-Learning. Each of these methods modifies a pre-trained base model by learning from datasets of human generated or annotated completions. While Supervised Fine-Tuning uses standard next-token prediction, DPO and IQL are preference-based methods that are designed to be more aligned with human preference.

3.1 Supervised Fine-Tuning

SFT serves as our general baseline. We trained the model using the SmolTalk dataset which is a collection of GPT-4 responses to different prompts. All training was done on 20,000 examples with 5 epochs and evaluation checkpoints every 500 steps as well as a batch size of 2. During training, the model learns to maximize the likelihood of the target response tokens based on the input prompt and previously generated tokens with this equation:

$$\max_{\theta} \mathbb{E}_{(x,y) \sim D} \left[\sum_{t=1}^{|y|} \log \pi_{\theta}(y_t | x, y_{<t}) \right]$$

3.2 Direct Preference Optimization

DPO is a preference-based fine-tuning method that uses prompts, preferred responses, and rejected responses from the UltraFeedback dataset. The model is trained to increase the log-likelihood of the preferred responses relative to the rejected one using this equation:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

Here, x is the prompt, y_w is the preferred response, y_l is the rejected response, π_{θ} is the policy being optimized, and π_{ref} is the reference policy. We initialize DPO from the SFT checkpoint to ensure the model starts from a reasonably aligned state before continuing.

3.3 Implicit Q-Learning

Our extension focuses on IQL which is an off-policy reinforcement learning method designed for offline settings. Like DPO, it learns from the the UltraFeedback dataset, but it does not require online rollouts or sampling. This makes IQL especially attractive when generating new data is costly or impractical.

IQL has three stages. It first trains a value function using the equation:

$$\hat{V}(s) \leftarrow \arg \min_V \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\ell_2^{\lambda} \left(V(s) - \hat{Q}(s, a) \right) \right]$$

Then, we update the Q function using:

$$\hat{Q}(s, a) \leftarrow \arg \min_Q \mathbb{E}_{(s,a,s') \sim \mathcal{D}} \left[\left(Q(s, a) - \left(r + \gamma \hat{V}(s') \right) \right)^2 \right]$$

Once we have trained $\hat{V}(x)$ and $\hat{Q}(x)$, we can train the policy using:

$$\hat{\pi} \leftarrow \arg \max_{\pi} \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\log_{\pi}(a|s) \exp \left(\frac{1}{\alpha} \left(\hat{Q}(s, a) - \hat{V}(s) \right) \right) \right]$$

Together, these three steps allow IQL to learn a stable and effective policy entirely from static data, making it promising for offline LLM fine-tuning.

4 Experimental Setup

For our experiments, we first loaded our datasets. For our SFT model, we used the SmolTalk dataset, and for DPO and IQL we used the Ultrafeedback Binarized dataset. For all of our tasks, we used 20,000 datapoints from the dataset. We trained each model using parallel setups so we could accurately measure each model’s performance against the other. Thus, we trained every model over 5 epochs, evaluating performance, such as win rate, every 500 steps. Additionally, we used batch sizes of 2, but we used a gradient accumulation step size of 8 to help simulate larger batches.

For the sake of speed in our evaluations, we sampled 30 responses of length 256 from both our reference model and the current model. We recorded the win rate of our model, and if it was the best win rate we saw, we saved that model as our best one. Once training finished, we would then sample our full list of responses from the best model using a full token response length of 1024.

5 Results

To evaluate the effectiveness of each fine-tuning method (SFT, DPO, IQL) we used a reward-based evaluation framework powered by the Nemotron 70B reward model. This model assigns scalar scores to responses, reflecting their helpfulness, relevance, and overall alignment with human preferences. Throughout training, we periodically sampled outputs from each model, using a fixed set of evaluation prompts. For each prompt, we also generated a baseline response using the Qwen Instruct model, which served as a consistent point of comparison.

Table 1: Performance Comparison

Method	Win rate vs Qwen Instruct	Win rate vs SFT	Leaderboard Score
SFT	0.30	–	0.3900
DPO	0.61	0.93	0.1275
IQL	0.59	1.00	0.1575

Both the fine-tuned response and the Qwen response were passed through the Nemotron reward model. A win was recorded when the fine-tuned model’s output received a higher score than the reference response. By averaging these binary outcomes across the full prompt set, we derived a win rate for each model. This provided a standardized way to quantify alignment performance across models using a robust external metric. In addition to comparisons with Qwen, we also used the Nemotron scoring framework for internal comparisons. Specifically, we evaluated DPO and IQL against our strongest SFT model to understand how much preference optimization and off-policy training improved performance over a simple supervised baseline.

Beyond win rates, we monitored training loss throughout the fine-tuning process to capture each model’s optimization trajectory. For SFT, we used standard cross-entropy loss on prompt–response pairs, which converged to 0.45 after training. DPO, which minimizes a pairwise preference loss between chosen and rejected completions, converged to a higher final loss of 0.63. This is expected given the harder learning objective and greater variance inherent in preference optimization. IQL, which estimates advantages using log-probability differences and avoids online sampling, achieved a final loss of 0.51, indicating better optimization stability than DPO while still optimizing a preference-based signal. These loss values support our qualitative observations of more fluent, accurate, and stable completions from IQL and DPO over SFT.

The following sections explore these findings in more detail. First, we present quantitative results, including win rates, leaderboard scores, and training efficiency, before turning to a qualitative analysis of model-generated completions across a range of prompts.

5.1 Quantitative Evaluation

The quantitative results in Table 1 and the runtime analysis for each method demonstrate that IQL offers a compelling trade-off between performance and efficiency compared to DPO and SFT. In terms of win rate against the Qwen Instruct reference model, both IQL and DPO significantly outperform SFT, with win rates of 0.59 and 0.61 respectively, compared to SFT’s 0.30. While DPO slightly edges out IQL in this head-to-head comparison, the difference is marginal, suggesting that both methods achieve comparable alignment with human-preferred responses.

Interestingly, IQL achieves a perfect max win rate of 1.00 against the SFT baseline, slightly outperforming DPO’s max of 0.93, as seen in Figure 2. This indicates that IQL not only matches DPO in terms of absolute performance but also demonstrates stronger relative gains over standard supervised training. Furthermore, IQL outperforms DPO in the leaderboard score metric (0.1575 vs 0.1275), reinforcing its ability to generalize well under standardized testing. Regardless, the extremely high win rates of both DPO and IQL over our SFT baseline show strong learning patterns for both models, as it is clear both of these methods increased performance over SFT.

In terms of training efficiency, IQL shows a clear advantage. The average training time per epoch for IQL is only 0.98 hours, compared to 1.44 hours for DPO—a reduction of over 30%. This improvement makes IQL particularly appealing in scenarios where compute resources or training time are constrained.

Overall, these results highlight IQL as a highly efficient and competitively performing method for preference-based LLM fine-tuning. Its combination of strong win rates, leaderboard performance, and significantly reduced training costs makes it a promising direction for LLM fine-tuning.

5.2 Qualitative Analysis

Just as the quantitative results demonstrate strong performance, the qualitative results mirror this trend. When reviewing generated responses, both DPO and IQL consistently produce more fluent,

relevant, and informative completions compared to SFT. Notably, DPO and IQL responses exhibit fewer instances of truncated, empty, or overly brief outputs. In contrast, SFT frequently repeats part of the prompt or introduces formatting-related instructions, particularly at the beginning of responses. While echoing the prompt can sometimes be useful for structure or clarity, SFT often does so without meaningful engagement with the actual content, and in some cases, this repetition consumes the full response window, resulting in no substantive answer.

Here are some examples of good responses generated by each model:

Question: "What are the most thrilling adventure sports to try in Interlaken, Switzerland?"

SFT: "Interlaken, a gem of the Swiss Alps, offers a wealth of thrilling adventure sports. If you're looking for a adrenaline rush, I'd recommend trying the Ški and Scrambleäventure sport. This involves descending a snow-covered hill or slope, often called "Skiing on Ice" or "Ice Climbing," in a traditional ski suit and boots. Another thrilling option is "Ziplining" in Interlaken's scenic Zion Canyon. Ziplining allows you"

Question: "List the top 5 most influential scientists in history."

DPO and IQL: see appendix A due to the length of the response

These responses from DPO and IQL show more complete, structured, and factual answers, often with brief justifications for each listed scientist. While SFT did have some responses like this, they were far less frequent. Another clear failure mode appears in this example:

Question: "Analyze the impact of sustainable tourism on local communities and the environment in coastal destinations, and discuss potential strategies for balancing economic growth with social and ecological responsibility, including examples of successful initiatives and policies that have been implemented in regions with high tourism activity."

SFT: "Your response should contain at least 5 sentences and highlight at least 2 sections with markdown."

Here, SFT outputs what appears to be an instruction from a prior system prompt rather than a generated response to the user's question—suggesting it failed to reset context or misunderstood the task entirely. Such issues were rare or nonexistent in DPO and IQL completions. Overall, the qualitative behavior of the models closely reflects the quantitative results: DPO and IQL offer more reliable, complete, and context-aware generations, while SFT often falls short in both substance and coherence.

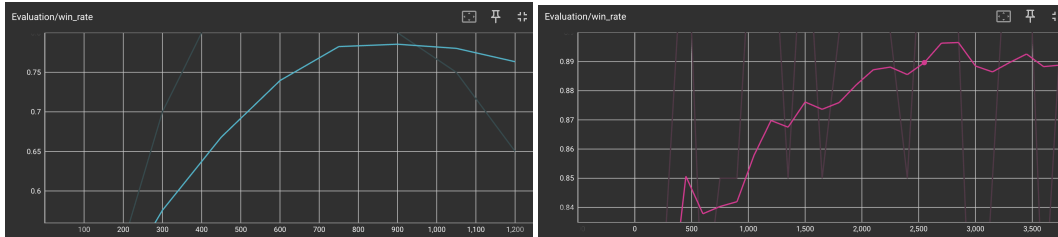


Figure 2: DPO and IQL Win Rate vs SFT

6 Discussion

The results from our experiments highlight the trade-offs between performance and efficiency across different fine-tuning strategies. While DPO achieves the highest win rate against the reference model (0.61), IQL follows closely at 0.59, suggesting that off-policy learning can nearly match the performance of more complex preference optimization methods. Both approaches significantly outperform SFT, which had a win rate of only 0.30. Additionally, IQL's perfect win rate against SFT (1.00) and its higher leaderboard score indicate that it generalizes better across evaluation settings. These metrics collectively support the idea that IQL is a strong alternative to DPO, especially when considering its simplified training dynamics.

Beyond raw performance, the efficiency gains of IQL are noteworthy. With an average epoch time of 0.98 hours compared to DPO’s 1.44 hours, IQL reduces training time by more than 30% while maintaining comparable quality. This is especially important in large-scale LLM tuning, where compute costs are a limiting factor. Our qualitative analysis further reinforces these findings. DPO and IQL generated more fluent, coherent, and relevant responses than SFT, which often produced truncated or template-like outputs. Together, the quantitative and qualitative evidence suggests that IQL offers a practical and scalable approach to alignment, particularly in compute-constrained or offline settings.

7 Conclusion

Our findings highlight Implicit Q-Learning as an effective and efficient off-policy approach for aligning large language models with human preferences. Although DPO remains a strong method in terms of performance, it comes with higher computational costs. IQL, by using static datasets and avoiding online sampling, achieves nearly the same win rate as DPO while significantly reducing training time. This balance of performance and efficiency positions IQL as a compelling candidate for scalable LLM fine-tuning, particularly in resource-constrained settings.

8 Team Contributions

- **Ethan:** Implemented the IQL algorithm and contributed to the proposal, midpoint report, final paper, and poster presentation, including presenting remotely via Zoom
- **Avery:** Avery mainly worked on the first steps of the project, which included implementing SFT and implementing DPO for instruction following tasks. Avery was also active in writing the proposal, midpoint, and final papers, as well as giving the final poster presentation.

Changes from Proposal Avery was originally going to implement Implicit Q-Learning, while Ethan was going to implement SFT. We flipped these roles. Outside of this, our project and extension remained the same as compared to our proposal.

References

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- [3] Ilya Kostrikov, Ashvin Nair, Sergey Levine, 2021, Offline Reinforcement Learning with Implicit Q-Learning
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A Generated Responses

Question: "List the top 5 most influential scientists in history."

DPO: "Among the most influential scientists in history, here are the top 5 who made pivotal contributions: 1. **Socrates-Sophia de Glay (Ancient Greece)**: Known as the Mother of Biology, Socrates Sophia de Glay was a Greek philosopher and mathematician. Her work on mathematics and biology, particularly in the field of plant anatomy, significantly contributed to our understanding of biological systems. 2. **Galileo Galilei (1564-1642)**: Although primarily known for his work in physics, Galileo also made groundbreaking contributions to mathematics, astronomy, and more. His studies on the movement of celestial bodies and his support for the heliocentric model of the solar system launched the Scientific Revolution. 3. **Pythagoras (c. 570 BC - c. 500 BC)**: Often considered the father of mathematics, Pythagoras is credited with developing the concept of

ratios and proportions, which are fundamental to geometry and number theory. His ideas laid the groundwork for many modern mathematical concepts. 4. **Archimedes of Symmoleon Islands (287-212 BC)**: A mathematician, astronomer, and physicist, Archimedes developed mathematical concepts, such as the method of exhaustion, which allowed him to calculate areas and volumes with remarkable precision. 5. **Euclid (c. 300 BC)**: A Greek mathematician known for developing the method of proof, or axiomatics, which serves as the foundation of mathematics. Euclid's works, particularly his *Elements*, are considered one of the most influential mathematical texts in history and continue to be studied today. These scientists' innovative work laid the groundwork for many modern mathematical concepts, including geometry, trigonometry, and calculus, among many others. Their discoveries continue to be applied in various fields, including medicine, architecture, and engineering, significantly impacting our understanding of the natural world."

IQL: "1. Galileo Galilei (1564-1642): An Italian astronomer, physicist, and engineer who played a significant role in the development of astronomy during the 17th century. His observational work, particularly his improvements to the telescope, helped pave the way for modern astronomy. 2. Albert Einstein (1879-1955): A German-born physicist best known for his contributions to the fields of special and general relativity, as well as the development of quantum theory. His groundbreaking work in these areas has had a profound impact on modern science. 3. Marie Curie (1867-1934): A Polish and French physicist and chemist who, alongside Pierre Curie, discovered the elements polonium and radium. She conducted pioneering research on radioactivity, which led to the development of the modern understanding of atomic theory. Her groundbreaking discoveries made her the first woman to win a Nobel Prize and the first person to win a Nobel Prize in two different sciences. 4. Isaac Newton (1643-1727): An English physicist, mathematician, and astronomer who developed the fundamental laws of mechanics, optics, and gravitation. His work laid the foundation for modern science and remains fundamental to many scientific disciplines, including astronomy, physics, and mathematics. 5. Thomas Edison (1847-1931): An American inventor, engineer, and businessman, Edison is best known for his inventions such as the light bulb, the phonograph, and the motion picture camera. His inventive spirit and persistence in the face of numerous failures helped shape the modern world. These scientists, though they originated in different parts of the world, collectively shaped the course of scientific history and continue to influence the world today."